

Question

Write a Java program to validate a mobile number and demonstrate user-defined and built-in exceptions.

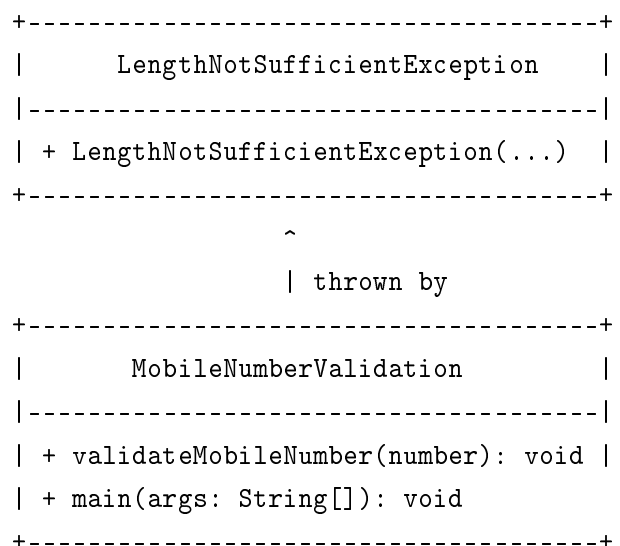
Steps for Execution

1. Save the code as `MobileNumberValidation.java`.
2. Compile the program.
3. Run `MobileNumberValidation`.
4. Enter a mobile number when prompted.

Algorithm

1. Read the mobile number as text.
2. Check that every character is a digit.
3. If the number has more than 10 digits, the array check reports an error.
4. If it has fewer than 10 digits, throw the custom length exception.
5. Otherwise, display that the number is valid.
6. Display that validation is complete.

UML Class Diagram



CODE: SRC

MobileNumberValidation.java

```
import java.util.Scanner;

// User-defined exception
class LengthNotSufficientException extends Exception {

    public LengthNotSufficientException(String message) {
        super(message);
    }
}

public class MobileNumberValidation {

    static void validateMobileNumber(String number)
        throws LengthNotSufficientException {

        try {
            // Check for characters other than digits
            for (int i = 0; i < number.length(); i++) {
                if (!Character.isDigit(number.charAt(i))) {
                    throw new NumberFormatException();
                }
            }

            // Check if length is greater than 10
            if (number.length() > 10) {
                int[] arr = new int[10];
                for (int i = 0; i < number.length(); i++) {
                    arr[i] = Character.getNumericValue(number.charAt(i));
                }
            }

            // Check if length is less than 10
            if (number.length() < 10) {
                throw new LengthNotSufficientException(
                    "Invalid Mobile Number - LengthNotSufficientException"
                );
            }

            System.out.println("Valid number");
        }
    }
}
```

```

    } catch (ArrayIndexOutOfBoundsException e) {

        System.out.println(
            "Invalid Mobile Number-ArrayIndexOutOfBoundsException"
        );

    } catch (NumberFormatException e) {

        System.out.println(
            "Invalid Mobile Number -NumberFormatException"
        );
    }
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter mobile number: ");
    String number = sc.nextLine();

    try {
        validateMobileNumber(number);
    } catch (LengthNotSufficientException e) {
        System.out.println(e.getMessage());
    } finally {
        sc.close();
        System.out.println("Validation completed.");
    }
}
}

```

Compilation and Execution

```

javac MobileNumberValidation.java
java MobileNumberValidation

```

Output

```

Enter mobile number: 9876543210
Valid number
Validation completed.

```